

Transformers

Physical Commonsense Reasoning on PIQA

NLP – Sacha Vogel

Preprocessing

DATA

[ybisk/piqa](#) into

training split	15'113
validation split	1'000
test split	1'838

FEATURE SELECTION

```
COL_GOAL = 'goal'  
COL_SOL1 = 'sol1'  
COL_SOL2 = 'sol2'  
COL_LABEL = 'label'
```

FILTER HTML ELEMENTS

Input / Output Format Transformers

TOKENIZATION

bert-base-uncased

truncate at 126 (99th percentile)

(OOM with max input length - 512)

TEXT EXAMPLE

['[CLS]', 'when', 'boiling', 'butter', ',', 'when', 'it', '"', 's', 'ready', ',', 'you', 'can',
'[SEP]', 'pour', 'it', 'onto', 'a', 'plate', '[SEP]']

Input / Output Format Transformers

TOKENIZATION

label	<i>0 or 1</i>
input1_attention_mask	<i>always 1</i>
input1_input_ids	<i>token ids</i>
input1_token_type_ids	<i>goal 0 / sol 1</i>
input2_attention_mask	<i>always 1</i>
input2_input_ids	<i>token ids</i>
input2_token_type_ids	<i>goal 0 / sol 1</i>

Input / Output Format Transformers

DATA LOADER

BATCH_SIZE = 32

pad to longest input sequence in batch

create tensor for each input

FINAL INPUT FORMAT

encode inputs into [CLS] vector (768 dim)

concatenate CLS1 & CLS2 into (1536 dim)

Input / Output Format Transformers

OUTPUT

2 logits

argmax of those logits $\longrightarrow$ predicted label *0 / 1*

SOL1 is correct if predicted label = 0

SOL2 is correct if predicted label = 1

Input / Output Format LLM

PROMPTS

system prompt – assign role

user prompt – including goal, sol1 and sol2

TOKENIZATION

Qwen2.5-3B-Instruct

apply_chat_template

→

<|system|>

system prompt

<|user|>

user prompt – including goal, sol1 and sol2

<|assistant|>

input_ids

attention_mask

Input / Output Format LLM

OUTPUT

answer is concatenated with input

split answer to output

decode

parse into predicted label

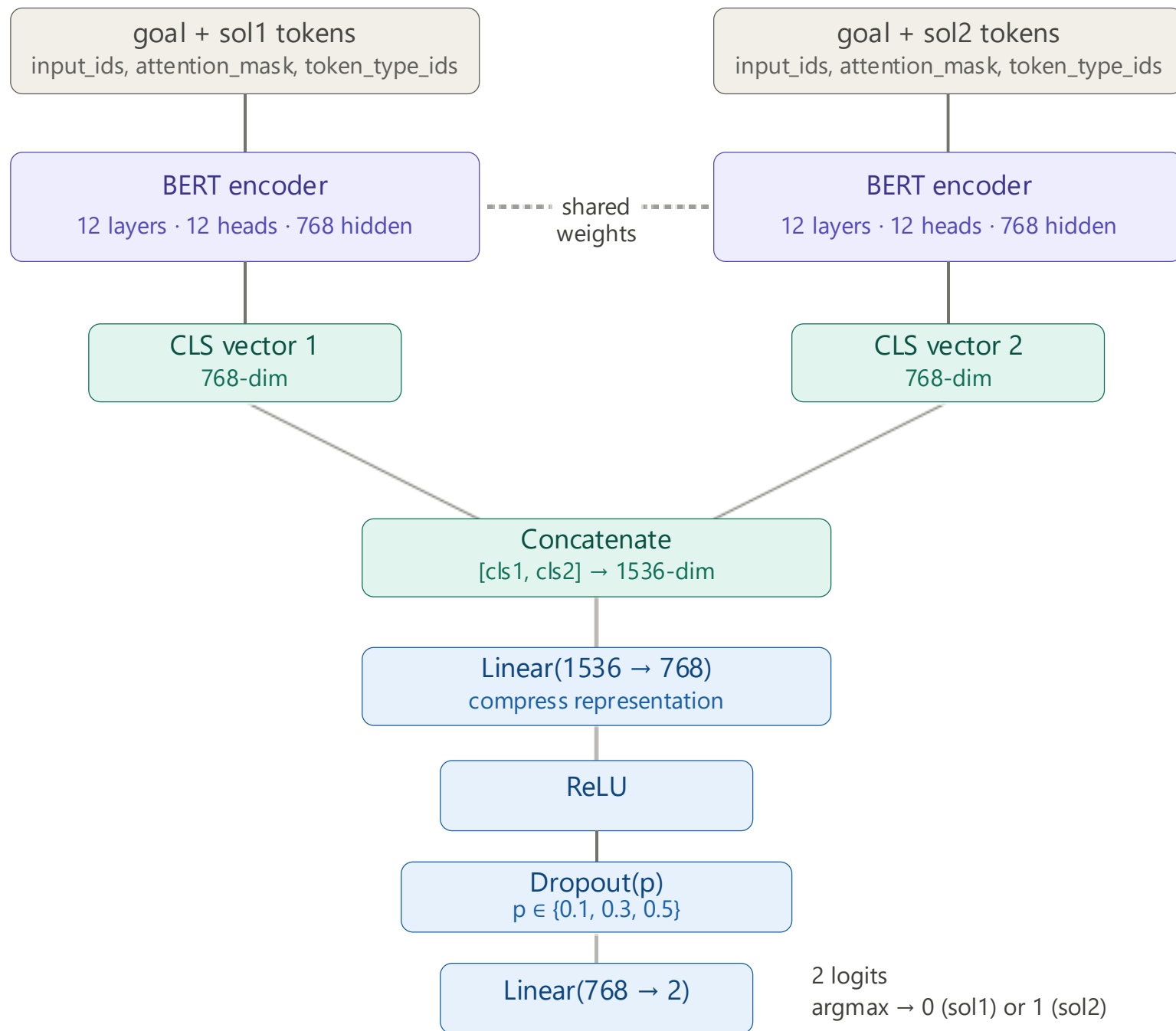
[... input tokens ..., 32, 198]

[32, 198]

tokenizer.decode([32, 198]) → "A"

A = 0 / B = 1 / default = -1

Network Architecture Transformers



Network Architecture LLM

“ARCHITECTURE”

Qwen2.5-3B-Instruct

3B parameters

Experiments

PRETRAINED TRANSFORMER

add **OneCycleLR** scheduler after low regularization

LLM

switched from **Qwen2.5-1.5B-Instruct** to **Qwen2.5-3B-Instruct**

try second prompt with specific role assignment

answer with «A» and «B» instead of «0» and «1» improved performance

Results

MODEL	TRAINING SET ACCURACY	VALIDATION SET ACCURACY	TEST SET ACCURACY
Random Initialized BERT	0.504	0.517	0.496
Pretrained BERT (bert-base-uncased)	0.956	0.668	0.656
LLM (Qwen2.5-3B-Instruct)	-	-	0.753

Interpretation

RANDOM INITIALIZED TRANSFORMER

always predicts sol1

never learned anything

are 110B params too much to learn anything for about 15'000 data records?

PRETRAINED TRANSFORMER

meaningful predictions

bad regularization

OneCycleLR scheduler improved test set acc. by 0.056

(pretrained weights are not destroyed)

Interpretation

LLM

solid predictions

Qwen2.5-**3B-Instruct** outperformed Qwen2.5-**1.5B-Instruct**

(by 0.143 on the same prompt)

KEY TAKEAWAY

models fail on near identical solutions

physical commonsense world knowledge in training data of LLM outperforms
task specific fine tuning

**Thank You For Your
Attention!**

Are There Any **Questions?**